

STEP 1: Create project folder

```
mkdir program-3  
cd program-3
```

STEP 2: Create Dockerfile

```
nano Dockerfile
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~$ mkdir program-3  
cd program-3  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ nano Dockerfile  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ cat Dockerfile  
FROM python:latest  
  
WORKDIR /app  
  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
COPY . .  
  
EXPOSE 5000  
  
CMD ["python", "app.py"]
```

STEP 3: Create app.py

```
nano app.py
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ nano app.py  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ cat app.py  
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    return "Hello from Simple Flask Docker!"  
  
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

STEP 4: Create requirements.txt

```
nano requirements.txt
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ nano requirements.txt  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ cat requirements.txt  
Flask==2.3.3
```

STEP 5: Verify project structure

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ ls  
app.py  Dockerfile  requirements.txt  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ docker build -t program-3 .
```

STEP 6: Build Docker image

`docker build -t program-3 .`

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ docker build -t program-3 .
[+] Building 61.1s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 197B                                0.0s
=> [internal] load metadata for docker.io/library/python:latest    3.8s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a 41.1s
=> => resolve docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a 0.0s
=> => sha256:3b700cf7651a764f8c33f574fe4f71be54f77cf626c3b2572e26493ca465e3b1 2.32kB / 2.32kB 0.0s
=> => sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 49.29MB / 49.29MB 10.2s
=> => sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 25.61MB / 25.61MB 5.8s
=> => sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9 67.78MB / 67.78MB 14.6s
=> => sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a 10.95kB / 10.95kB 0.0s
=> => sha256:1edf53fea0281306069b0c05371a2ad9c5ac12e7474dc0cd6c4902c7eb33a9c9 6.49kB / 6.49kB 0.0s
=> => sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 235.98MB / 235.98MB 25.4s
=> => sha256:b1741a62ccced4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 6.08MB / 6.08MB 11.8s
=> => extracting sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 3.6s
=> => sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB 16.7s
=> => extracting sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 1.6s
=> => sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd8d468d 249B / 249B 15.5s
=> => extracting sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9 5.4s
=> => extracting sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 13.3s
=> => extracting sha256:b1741a62ccced4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 0.4s
=> => extracting sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 1.5s
=> => extracting sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd8d468d 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 93B                                       0.0s
=> [2/5] WORKDIR /app                                              0.9s
=> [3/5] COPY requirements.txt .                                    0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt        14.6s
```

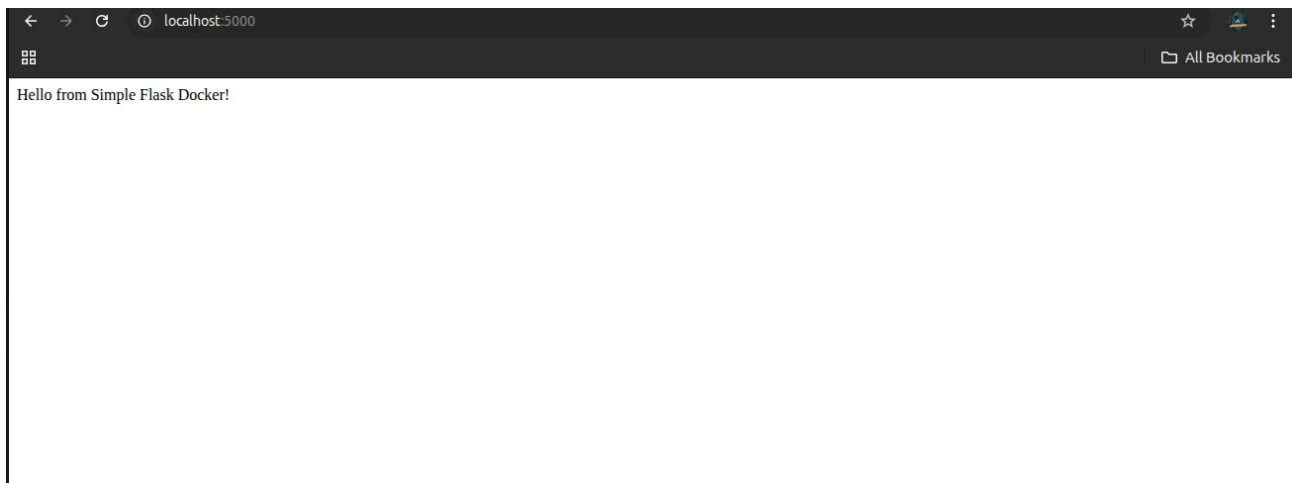
STEP 7: Run Docker container

`docker run -p 5000:5000 program-3`

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-3$ docker run -p 5000:5000 program-3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [07/Jan/2026 05:32:52] "GET / HTTP/1.1" 200 -
```

STEP 8: Test in browser

Open **Chrome** and go to:



STEP 9: Remove container & image

List containers

```
docker container ls -a
```

Stop container

```
docker container stop
```

Remove container

```
docker container rm
```

Remove image

```
docker image rm program-3
```

```
1RV24MC087_SSDhanush@dhhanush-Latitude:~/program-3$ docker container rm 65bf4445c50a
65bf4445c50a
1RV24MC087_SSDhanush@dhhanush-Latitude:~/program-3$ docker image rm program-3
Untagged: program-3:latest
Deleted: sha256:718affb4a30513b5fc84aa31aa6e2203068ca9618139e86e8dd9a812645d5355
1RV24MC087_SSDhanush@dhhanush-Latitude:~/program-3$ docker images
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE
1RV24MC087_SSDhanush@dhhanush-Latitude:~/program-3$
```